

# An explicit positive-definite counterexample to half-line q-permanent monotonicity

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## Abstract

We exhibit a rational real symmetric positive-definite matrix of order four whose inversion-weighted q-permanent decreases between  $q=49$  and  $q=50$ . Exact principal minors establish positive definiteness, and a six-term permutation computation gives the complete polynomial. The counterexample disproves da Fonseca's positive-definite half-line extension, independently of the convention at its left endpoint. The actual matrix and q-permanent counterexample are completely formalized in Lean.

## 1 The half-line conjecture

For a real matrix  $A = (a_{ij})$  of order  $n$ , define

$$P_q(A) = \sum_{\sigma \in S_n} q^{\text{inv}(\sigma)} \prod_{i=1}^n a_{i, \sigma(i)}, \quad \text{inv}(\sigma) = \#\{(i, j) : i < j, \sigma(i) > \sigma(j)\}.$$

The index ordering is part of the definition. Da Fonseca's half-line extension asks whether, for a real symmetric positive-definite non-diagonal matrix, there is a left endpoint at or below  $-1$  from which this function is strictly increasing up to  $+\infty$ . His 2018 formulation uses an endpoint strictly below  $-1$ . A decrease between 49 and 50 refutes both versions, independently of the left endpoint convention.

## 2 The exact counterexample

**Theorem.** The rational matrix

$$A = \begin{pmatrix} 1 & -99/100 & 0 & 1/500 \\ -99/100 & 1 & 0 & 1/8 \\ 0 & 0 & 1 & 0 \\ 1/500 & 1/8 & 0 & 1 \end{pmatrix}$$

is positive definite, and  $P_{49}(A) > P_{50}(A)$ . Consequently it disproves the positive-definite half-line conjecture.

**Proof.** Its successive leading principal minors are

$$1, \quad \frac{199}{10000}, \quad \frac{199}{10000}, \quad \frac{59}{15625}.$$

They are strictly positive, so Sylvester’s criterion proves positive definiteness. A permutation can contribute only if it fixes index 3. The six permutations of indices  $\{1, 2, 4\}$  give the following complete list.

| Permutation | Inversions | Monomial contribution |
|-------------|------------|-----------------------|
| Identity    | 0          | 1                     |
| (12)        | 1          | $(9801/10000)q$       |
| (24)        | 3          | $(1/64)q^3$           |
| (14)        | 5          | $(1/250000)q^5$       |
| (124)       | 4          | $-(99/400000)q^4$     |
| (142)       | 4          | $-(99/400000)q^4$     |

Thus exact enumeration, retaining all four index positions, gives

$$P_q(A) = 1 + \frac{9801}{10000}q + \frac{1}{64}q^3 - \frac{99}{200000}q^4 + \frac{1}{250000}q^5.$$

Evaluation yields

$$P_{49}(A) = \frac{81807513}{500000}, \quad P_{50}(A) = \frac{7969}{50}, \quad P_{49}(A) - P_{50}(A) = \frac{2117513}{500000} > 0.$$

Since  $49 < 50$ , the theorem follows. No numerical optimization, approximate eigenvalue or derivative test is used.  $\square$

### 3 Scope and reproduction

This is the positive-definite half-line conjecture, not Bapat’s original monotonicity conjecture restricted to  $[-1, 1]$ . Deleting the isolated identity index or moving the nonzero triangle to three consecutive indices would change the inversion statistic and the displayed polynomial. The checker `verify_exact.py` independently enumerates all permutations and evaluates every determinant and polynomial value in rational arithmetic. The accompanying Lean proof works with the actual matrix positive-definiteness definition and the actual inversion-weighted permutation sum, rather than assuming a surrogate polynomial identity. The separate minimum-dimension and arbitrary- $t > 1$  extensions in the earlier research note are outside this preprint and its formalization claims.

### 4 References

1. C. M. da Fonseca, *The mu-permanent of a tridiagonal matrix, orthogonal polynomials, and chain sequences*, Linear Algebra and its Applications **432** (2010), 1258–1266, Conjecture 1, p. 1259. <https://doi.org/10.1016/j.laa.2009.10.036>.
2. C. M. da Fonseca, *The mu-permanent revisited*, arXiv:1804.02231v1 (2018), Conjecture 2. <https://arxiv.org/abs/1804.02231v1>.
3. L. Mitchell, *A note on Bapat’s q-permanent conjecture*, Operators and Matrices **14** (2020), 915–919, especially the remark on pp. 917–918 distinguishing the positive-definite conjecture from its singular extension. <https://doi.org/10.7153/oam-2020-14-56>.

## 5 Formal verification and reproduction

The principal theorem has a complete Lean proof. The precise scope is: Only the displayed 4-by-4 real rational positive-definite counterexample and its exact decrease between  $q=49$  and  $q=50$ . Minimal dimension and the all- $t>1$  family from the older research note are outside this paper. The public proof source snapshot is fixed at commit `02af7db99db789fecb0d5f5f8f37b1a4e40762a0` in <https://github.com/mxym/math>. The proof packages, with their own dependency locks, audit records and reproduction instructions, are:

- `formalizations/q-permanent-halfline-counterexample`.

The main formal entry points are:

- `QPermanentHalfline.explicit_counterexample`.
- `QPermanentHalfline.halfLineConjecture_false`.

Lean is fixed to version 4.34.1, compiler commit `5045d0056413266e57c625dcd7c365b10e377c52`; Mathlib is fixed to `d13f23b723b8a846827a245b89c10fc7d3f11612`. The audited mathematical roots depend only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The archived audits state their actual compilation and replay scopes; publication of this typeset version does not imply a new execution of every previous empty-kernel replay. The companion README maps each printed principal result to its formal theorem and explains how to repeat verification.

## Research and preparation disclosures

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